

The Diffie-Hellman-Merkle key exchange protocol









A, p, g

A thick, dark gray horizontal arrow pointing to the right, spanning the width of the image. It is positioned below the text A, p, g .



1. Generates public numbers p and g such that g is co-prime to $p-1$
2. Generates a secret number a
3. Sends $A = g^a \bmod p$ to Bob

1. Generates a secret number b
2. Sends $B = g^b \bmod p$ back to Alice
3. Calculates the key $K = A^b \bmod p$

4. Calculates the key $K = B^a \bmod p$

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Diffie-Hellman-Merkle in practice

- g is small (either 3, 5 or 7 and fixed in practice)
 - p is at least 2048 bits (and fixed in practice)
 - private keys a and b are 2048 bits as well
- ➔ So the public values A and B
and the master key k are 2048 bits
- ➔ Use k to derive an AES key using a Key Derivation Function
(usually HKDF - the HMAC-based Extract-and-Expand key derivation function)